

Bounding the Infinite-Bagged RSM Remainder under Common Loading

A randomized-linear-algebra/coherence argument

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1 Purpose

This note gives a clean way to bound the infinite-bagged RSM remainder

$$R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}},$$

under the common-loading heteroskedastic forecast-error structure. The goal is not to assume a functional form such as C/k^2 , but to show when such a rate is theoretically justified. The randomized-linear-algebra (RandNLA) idea enters through leverage coherence: uniform random subspace sampling is close to leverage sampling only when the leverage/oracle weights are not too concentrated. When leverage is highly concentrated, the safe bound is the Jensen bound $0 \leq R_k^{\text{bag},\infty} \leq Q_k^{\text{sub}} - Q^{\text{opt}}$.

2 Common-loading setup and exact remainder

Assume the common-loading covariance structure

$$\Sigma_e = q \mathbf{u} \mathbf{u}' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2), \quad \tau_i = \sigma_i^{-2} > 0.$$

Let

$$A_m = \sum_{i=1}^m \tau_i = m\bar{\tau}, \quad \ell_i = \frac{\tau_i}{A_m}, \quad \sum_{i=1}^m \ell_i = 1.$$

Under common loading the full oracle combination weight is $w_i^* = \tau_i/A_m = \ell_i$, so the oracle weights are exactly the normalized precision scores — the link to leverage-score sampling. For a subset S of size k , define the sampled leverage mass $L_S = \sum_{j \in S} \ell_j = A_S/A_m$, $A_S = \sum_{j \in S} \tau_j$. The embedded subset weight is $v_{S,i} = \mathbf{1}\{i \in S\} \tau_i/A_S = \mathbf{1}\{i \in S\} \ell_i/L_S$, so the infinite-bagged weight is $\bar{v}_{k,i} = \mathbb{E}_S[\mathbf{1}\{i \in S\} \ell_i/L_S]$. Since $\bar{v}_k' \mathbf{u} = 1$, the exact excess-risk identity is

$$R_k^{\text{bag},\infty} = (\bar{v}_k - w^*)' \Sigma_e (\bar{v}_k - w^*) = \sum_{i=1}^m \frac{(\bar{v}_{k,i} - w_i^*)^2}{\tau_i}.$$

Because $w_i^* = \ell_i$ and $\tau_i = A_m \ell_i$,

$$R_k^{\text{bag},\infty} = \frac{1}{A_m} \sum_{i=1}^m \ell_i \left(\frac{\bar{v}_{k,i}}{\ell_i} - 1 \right)^2. \quad (1)$$

This is the RandNLA form: $R_k^{\text{bag},\infty}$ is a weighted squared error from estimating the leverage/oracle vector by uniform subset sampling.

3 The coherence parameter

Define the leverage coherence and normalized precision dispersion

$$\kappa = m \max_i \ell_i = \frac{\tau_{\max}}{\bar{\tau}}, \quad h_i = m\ell_i - 1 = \frac{\tau_i - \bar{\tau}}{\bar{\tau}}, \quad V_\delta = \frac{1}{m} \sum_i h_i^2 = m \sum_i \left(\ell_i - \frac{1}{m} \right)^2.$$

Low coherence means no coordinate dominates the oracle weight; high coherence means a few forecasts carry a large fraction of it.

4 Low-coherence/balanced-leverage bound

Assumption 1 (Balanced leverage). *There exist constants $0 < c \leq C < \infty$ such that, for all m and all i , $c/m \leq \ell_i \leq C/m$, equivalently $c \leq \tau_i/\bar{\tau} \leq C$.*

Let $p = k/m$. Conditional on $i \in S$, write $L_{i,R} = \ell_i + \sum_{j \in R} \ell_j$, where R is a size- $(k-1)$ subset drawn uniformly without replacement from $\{1, \dots, m\} \setminus \{i\}$. Then $\bar{v}_{k,i}/\ell_i = p \mathbb{E}_R[1/L_{i,R}]$, with conditional mean

$$\mu_i = \mathbb{E}_R[L_{i,R}] = \ell_i + (k-1) \frac{1 - \ell_i}{m-1} = p + \frac{m-k}{m-1} \left(\ell_i - \frac{1}{m} \right), \quad z_k = \frac{1}{k} - \frac{1}{m} = \frac{m-k}{km}.$$

Lemma 1 (Reciprocal expansion with explicit remainder). *For each i , $\mathbb{E}_R[1/L_{i,R}] = 1/\mu_i + \mathbb{E}_R[(L_{i,R} - \mu_i)^2/(L_{i,R}\mu_i^2)]$, hence*

$$\frac{\bar{v}_{k,i}}{\ell_i} - 1 = \left(\frac{p}{\mu_i} - 1 \right) + p \mathbb{E}_R \left[\frac{(L_{i,R} - \mu_i)^2}{L_{i,R}\mu_i^2} \right].$$

Proof. Use the exact identity $1/x = 1/\mu - (x - \mu)/\mu^2 + (x - \mu)^2/(x\mu^2)$ with $x = L_{i,R}$, $\mu = \mu_i$; the first-order term vanishes since $\mathbb{E}_R[L_{i,R} - \mu_i] = 0$. $\square$

The first-order term is governed by deviations of ℓ_i from $1/m$; the second is a finite-population ratio-estimator bias.

Proposition 1 (Coherence-controlled RandNLA bound). *Under balanced leverage, for $2 \leq k < m$ and $m \geq 4$, there is a constant $K_{c,C} < \infty$, depending only on c, C , with*

$$R_k^{\text{bag},\infty} \leq \frac{K_{c,C}}{m\bar{\tau}} (V_\delta + V_\delta^2) z_k^2. \quad (2)$$

If V_δ is bounded this is $R_k^{\text{bag},\infty} = O(V_\delta z_k^2/(m\bar{\tau}))$, a $1/k^2$ envelope for $k \ll m$.

Proof sketch. Balanced leverage gives $L_{i,R} \geq cp$ and $\mu_i \geq cp$. Then $|p/\mu_i - 1| = |p - \mu_i|/\mu_i \leq K_c |h_i| z_k$. The SRSWOR variance $\text{Var}_R(L_{i,R}) \leq \frac{(k-1)(m-k)}{m(m-1)(m-2)} V_\delta$ controls the ratio-bias term by $K_c V_\delta z_k$, so $|\bar{v}_{k,i}/\ell_i - 1| \leq K_c z_k (|h_i| + V_\delta)$. Plugging into (1) and using $\ell_i \leq C/m$, $\sum_i \ell_i h_i^2 \leq C V_\delta$, gives (2). $\square$

Remark 1. *Uniform subset sampling is close to leverage sampling when leverage scores are balanced; the bias of uniform infinite bagging is then second order in the sampling distortion, so the excess risk has a z_k^2 envelope. This is the theoretical source of the cube-root rule in the low-coherence common-loading case. The envelope is an upper bound; the matching lower bound is supplied in Section 8.*

5 Using the low-coherence bound to choose k

Let $A = \sigma_\varepsilon^2 + Q^{\text{opt}} = \sigma_\varepsilon^2 + q + 1/A_m$. The infinite-bagged loss is $L_k^{\text{bag},\infty} = (A + R_k^{\text{bag},\infty}) \frac{T-1}{T-k}$. Under the low-coherence envelope $R_k^{\text{bag},\infty} \lesssim C_R z_k^2$, $C_R \asymp V_\delta/(m\bar{\tau})$, and $z_k \approx 1/k$ for $k \ll m$, the first-order condition $R'_k(T-k) + (A + R_k) = 0$ becomes $Ak^3 + 3C_R k - 2C_R T = 0$, so for large T

$$k^* \approx \left(\frac{2C_R T}{A} \right)^{1/3} \propto T^{1/3}. \quad (3)$$

6 High-coherence case: the safe Jensen bound

If leverage is highly concentrated the constants in (2) explode; this is the main economic case (a dominant high-precision forecast). Then do not use a z_k^2 bound unless verified empirically; use the exact Jensen sandwich.

Proposition 2 (High-coherence fallback). *For any positive precision vector and any $1 \leq k \leq m$,*

$$0 \leq R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}} \leq Q_k^{\text{sub}} - Q^{\text{opt}} = \mathbb{E}_S[1/A_S] - \frac{1}{A_m}. \quad (4)$$

Proof. Lower bound: w^* minimizes $w' \Sigma_e w$ over sum-to-one weights and $\bar{v}_k$ is feasible. Upper bound: Jensen, $Q_k^{\text{bag},\infty} = (\mathbb{E}_S v_S)' \Sigma_e (\mathbb{E}_S v_S) \leq \mathbb{E}_S[v_S' \Sigma_e v_S] = Q_k^{\text{sub}}$. Subtract Q^{opt} ; under common loading $Q(S) = q + 1/A_S$, $Q^{\text{opt}} = q + 1/A_m$. $\square$

Remark 2. *The high-coherence bound is first-order: $Q_k^{\text{sub}} - Q^{\text{opt}} \approx C_{\text{sub}}(1/k - 1/m)$ is closer to z_k than z_k^2 . This is why the effective exponent drifts from 2 toward 1 when a few high-precision forecasts dominate.*

7 Choosing k in the high-coherence case

Minimize the conservative upper bound $L_k^{\text{bag},\infty} \leq (A + Q_k^{\text{sub}} - Q^{\text{opt}}) \frac{T-1}{T-k}$. With the leading approximation $Q_k^{\text{sub}} - Q^{\text{opt}} \approx C_{\text{sub}}/k$ this returns square-root behavior $k^* \asymp (C_{\text{sub}} T/A)^{1/2} \propto T^{1/2}$.

8 A distribution-free two-sided bracket

The previous bounds are one-sided unless coherence is controlled. Here is a two-sided bracket that needs *no* coherence or distributional assumption, built only from the unconditional average-subset quantity $E := \mathbb{E}_S[1/A_S]$.

Write $g_i(k) = k \mathbb{E}_S[1/A_S \mid i \in S]$. Using $\bar{v}_{k,i} = \tau_i g_i(k)/m$, the exact remainder (1) becomes

$$R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i - \frac{1}{\bar{\tau}} \right)^2 = \frac{1}{m^2} \sum_{i=1}^m \tau_i g_i^2 - \frac{1}{A_m}. \quad (5)$$

Two *exact* linear identities hold for $\{g_i\}$:

$$\sum_{i=1}^m \tau_i g_i = m \quad (\text{conservation law}), \quad \sum_{i=1}^m g_i = km E \quad \left(\text{since } \frac{1}{m} \sum_i g_i = kE \right). \quad (6)$$

Proposition 3 (Closed-form two-sided bracket). *For $1 \leq k \leq m$,*

$$\frac{(k \mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i \sigma_i^2 - m/\bar{\tau}} \leq R_k^{\text{bag}, \infty} \leq \mathbb{E}_S[1/A_S] - \frac{1}{A_m}. \quad (7)$$

The denominator $\sum_i \sigma_i^2 - m/\bar{\tau} = \sum_i 1/\tau_i - m^2/A_m \geq 0$ by Cauchy-Schwarz, with equality iff the τ_i are equal (in which case $R_k^{\text{bag}, \infty} = 0$).

Proof. The upper bound is Proposition 2. For the lower bound, minimize $\sum_i \tau_i g_i^2$ over all vectors satisfying the two identities (6). In the inner product $\langle x, y \rangle_\tau = \sum_i \tau_i x_i y_i$ these read $\langle g, \boldsymbol{\iota} \rangle_\tau = m$ and $\langle g, c \rangle_\tau = kmE$ with $c_i = 1/\tau_i$. Orthogonalizing c against $\boldsymbol{\iota}$ ($\|\boldsymbol{\iota}\|_\tau^2 = A_m$, $\langle c, \boldsymbol{\iota} \rangle_\tau = m$, $\|c\|_\tau^2 = \sum_i 1/\tau_i$),

$$\min \sum_i \tau_i g_i^2 = \frac{m^2}{A_m} + \frac{(kmE - m^2/A_m)^2}{\sum_i 1/\tau_i - m^2/A_m} = \frac{m}{\bar{\tau}} + \frac{m^2(kE - 1/\bar{\tau})^2}{\sum_i 1/\tau_i - m/\bar{\tau}}.$$

Since the true g satisfies the identities, $\sum_i \tau_i g_i^2$ is at least this minimum. Substituting into (5) and using $\frac{1}{m^2} \cdot \frac{m}{\bar{\tau}} = \frac{1}{A_m}$, the oracle term cancels and (7) follows. $\square$

Remark 3 (Tightness and a hierarchy). *The Gram lower bound is exact whenever $g \in \text{span}\{\boldsymbol{\iota}, 1/\tau\}$, which holds for any profile with at most two distinct precision values (then g takes two values, fully pinned by the two identities). For continuous profiles g leaves that span only slightly, and the bound is near-exact numerically (within a few percent). Adding a third exact identity, e.g. $\sum_i \tau_i^2 g_i = m \mathbb{E}_S[\sum_{i \in S} \tau_i^2 / A_S]$, makes the bound exact for three-group profiles, yielding a convergent hierarchy.*

Both sides of (7) can be written as powers of z_k using the exact moments $\mathbb{E}_S[A_S] = k\bar{\tau}$ and $\text{Var}_S(A_S) = k \frac{m-k}{m(m-1)} V$, $V = \sum_i (\tau_i - \bar{\tau})^2$, together with $\mathbb{E}_S[1/A_S] = \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + \dots$.

Corollary 1 (Matching z_k forms). *To leading order, with $B = 1/\bar{\tau}$, $H = \bar{\sigma}^2 \bar{\tau}$, and $V_\delta = \text{Var}(\tau)/\bar{\tau}^2$,*

$$\underbrace{\frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2}_{\text{lower}} \lesssim R_k^{\text{bag}, \infty} \lesssim \underbrace{\frac{1}{\bar{\tau}} z_k}_{\text{upper}}. \quad (8)$$

In the small-heterogeneity limit $H - 1 \rightarrow V_\delta$, so the lower constant collapses to $V_\delta/(m\bar{\tau})$, matching the Hájek/ $\alpha = 2$ leading term.

Proof. Upper: $\mathbb{E}_S[1/A_S] - 1/A_m \approx \frac{1}{k\bar{\tau}} - \frac{1}{m\bar{\tau}} = z_k/\bar{\tau}$. Lower: $kE - 1/\bar{\tau} \approx \text{Var}_S(A_S)/(k^2 \bar{\tau}^3) = Vz_k/((m-1)\bar{\tau}^3)$; squaring and dividing by $\sum_i \sigma_i^2 - m/\bar{\tau} = mB(H-1)$, with $V = m\bar{\tau}^2 V_\delta$, gives the stated coefficient. $\square$

Since $0 < z_k < 1$, we have $z_k^2 < z_k$: the higher power is the *floor* and the lower power the *ceiling*, so the ordering in (8) is the natural one. The bracket holds for any $R_k \sim z_k^\alpha$ with $\alpha \in [1, 2]$, because $z_k^2 \leq z_k^\alpha \leq z_k$.

Corollary 2 (k^* bracket). *Substituting each z_k -form into $L_k = (A + R_k^{\text{bag}, \infty})^{\frac{T-1}{T-k}}$ with $z_k \approx 1/k$: the lower form $c_L z_k^2$ gives $Ak^3 + 3c_L k - 2c_L T = 0$, i.e. $k^* = \Theta(T^{1/3})$; the upper form Bz_k gives $Ak^2 + 2Bk - BT = 0$, i.e. $k^* = \Theta(T^{1/2})$. Hence*

$$\Theta(T^{1/3}) \leq k^* \leq \Theta(T^{1/2}),$$

the rate bracket corresponding to $k^ \propto T^{1/(\alpha+1)}$ with $\alpha \in [1, 2]$.*

9 Summary

1. Under common loading the oracle weights are leverage scores, $w_i^* = \ell_i = \tau_i/A_m$, and uniform RSM is a self-normalized uniform subset estimator of that vector.
2. If leverage is balanced, $R_k^{\text{bag},\infty} \leq \frac{K_{c,C}}{m\tau}(V_\delta + V_\delta^2)z_k^2$ (a $1/k^2$ envelope $\Rightarrow$ cube-root k^*).
3. If leverage is concentrated, use the Jensen fallback $0 \leq R_k^{\text{bag},\infty} \leq \mathbb{E}_S[1/A_S] - 1/A_m$ ($\Rightarrow$ square-root k^*).
4. **Distribution-free:** the Gram lower bound and the Jensen upper bound give the two-sided bracket (7), equivalently $c_L z_k^2 \lesssim R_k^{\text{bag},\infty} \lesssim B z_k$ and $\Theta(T^{1/3}) \leq k^* \leq \Theta(T^{1/2})$ — no coherence assumption. The lower bound is exact for ≤ 2 distinct precision values and near-exact otherwise.
5. The RandNLA/coherence story is the special case in which the two ends of the bracket pinch together ($\alpha \rightarrow 2$, low coherence) or separate maximally ($\alpha \rightarrow 1$, high coherence).